

Class 10 Science - Chapter 12

Magnetic Effects of Electric Current (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Ch11 + Class 6-8 ki yeh cheezein pehle pakki karo.

1. Magnet ka basic (Class 6)

Magnet ke do pole: North (N) aur South (S).

Same pole REPEL (door bhagaate), opposite pole ATTRACT (paas khinchte).

"Like poles repel, unlike poles attract."

Magnet ke around ek area hota jaha uska asar hota = MAGNETIC FIELD.

Compass needle (chumbak ki sui) hamesha North-South dikhati.

2. Current aur Circuit (Ch11 revision)

Current = charge ka flow (ampere). Conventional current +ve se -ve.

Conductor (taar) me current behne par uske around kuch hota -> isi chapter ka main idea (magnetic effect).

Solenoid = taar ko spring/coil jaise lapet kar banaya gaya.

3. Direction batane ke tareeke

3D direction: (dot .) = current/field BAAHAR aapki taraf;

(cross x) = ANDAR (paper ke andar) jaa rahi.

Perpendicular = 90 degree par. Right angle = 90 degree.

4. Maths/Geometry basics

Circle ke concentric (ek hi center wale) circles ka idea.

Direction (left/right/up/down) clearly samajhna - Fleming rules ke liye.

5. AC vs DC (idea)

DC (Direct current) = ek hi disha me (cell/battery se).

AC (Alternating current) = disha badalti rehti (ghar ka supply, India me 50 Hz matlab 1 second me 50 baar direction badalti -> 100 baar polarity change).

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

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[GREEN = board exam me baar-baar poocha jaata hai]

1. Magnetic Field aur Field Lines [IMPORTANT]

Oersted ne dekha: current wale taar ke paas compass sui ghoom jaati -> matlab current MAGNETIC FIELD banati hai.

Magnetic field = woh area jaha magnet/current ka magnetic asar hota.

Field ek VECTOR hai (magnitude + direction dono).

Magnetic field LINES ki properties:

- Bahar N se S ki taraf, magnet ke andar S se N (closed loops).
- Kabhi ek doosre ko CROSS nahi karti (kyunki ek point par 2 direction possible nahi).
- Paas-paas lines = strong field; door-door = weak field.

2. Field due to Current in a Straight Conductor [IMPORTANT]

Seedhe taar ke around magnetic field = concentric (ek center wale) CIRCLES.

Current jyada -> field strong; taar se door -> field weak.

RIGHT HAND THUMB RULE: right haath ka angootha CURRENT ki direction me; ungliyon ka curl (mudna) = magnetic field ki direction.

3. Field due to Solenoid + Electromagnet [VERY IMPORTANT]

Solenoid (coil) me current -> field bilkul BAR MAGNET jaisa.

- Ek sira N pole, doosra S pole. Andar field uniform (parallel lines).

ELECTROMAGNET = solenoid ke andar soft iron core daal do -> strong temporary magnet. Current band -> magnetism khatam (isliye "temporary").

(Soft iron core use hota kyunki jaldi magnetise + jaldi demagnetise hota.)

4. Force on Current-Carrying Conductor [TOP EXAM AREA]

Magnetic field me rakhe current wale taar par FORCE lagta (push).

Force max jab current field ke PERPENDICULAR (90 par) ho.

FLEMING'S LEFT HAND RULE (force/motion ke liye):

Left haath - 3 ungli 90 par phailao:

- First finger (tarjani) = Field (B)
- Centre finger (madhya) = Current (I)
- Thumb (angootha) = Force/motion (F)

Memory: "FBI" - thumb=Force, first=B(field), centre=I(current).

Yeh ELECTRIC MOTOR ka base hai.

5. Electric Motor [TOP EXAM AREA]

Electric motor = electrical energy ko MECHANICAL (ghoomne wali) energy me badle.
Working: magnetic field me coil me current -> dono side opposite force (Fleming left hand) -> coil GHoomTI.
SPLIT RING / COMMUTATOR = har aadhe chakkar me current ki direction PALAT-ta -> coil ek hi disha me ghoomti rehti.
Brushes = current ko coil tak pahunchate. Use: fan, mixer, pump, washing machine.

6. Electromagnetic Induction (EMI) [TOP EXAM AREA]

Faraday: badalti (changing) magnetic field se coil me current PAIDA hoti (bina battery) -> ELECTROMAGNETIC INDUCTION.
- Magnet ko coil ke paas hilao (ya coil ko) -> galvanometer me current dikhta.
- Jitni tezi se movement, utni jyada current.
FLEMING'S RIGHT HAND RULE (induced current ke liye):
Right haath - thumb=motion, first=field, centre=induced current.
Memory: motor=LEFT (force), generator/induction=RIGHT.

7. Electric Generator (Dynamo) [VERY IMPORTANT]

Generator = MECHANICAL energy ko ELECTRICAL energy me badle (motor ka ulta).
Working: coil ko magnetic field me ghumao -> EMI se current paida.
AC generator: SLIP RINGS use -> alternating current (AC) deta.
DC generator: SPLIT RING (commutator) use -> ek direction current (DC).
Ghar ka supply AC hota (India 220V, 50 Hz).

8. Domestic Electric Circuit + Safety [IMPORTANT]

Ghar me 3 taar aate:
- LIVE (red/brown) - +ve, current laata.
- NEUTRAL (black/blue) - current wapas le jaata.
- EARTH (green) - safety, leakage current ko zameen me bhej deta.
Earthing: metal body ko earth wire se jodte -> agar live taar body ko chhoo jaaye to current zameen me chala jaaye, insaan ko shock na lage.
Short circuit: live aur neutral seedha touch -> achaanak bahut current -> fuse pighal kar circuit tod deta.
Overloading: ek hi socket par bahut appliances -> jyada current -> aag ka khatra.
FUSE live wire me lagta (kam m.p. taar) - safety device.

9. Quick Revision - One Liners

- Current -> magnetic field (Oersted). Field lines cross nahi karti.
- Right hand thumb rule (field around wire); solenoid = bar magnet.
- Fleming LEFT hand = force (MOTOR). Fleming RIGHT hand = induced current (GENERATOR/EMI).
- Motor: electrical->mechanical (split ring). Generator: mechanical->electrical.
- Ghar: live/neutral/earth; earthing+fuse = safety; short circuit + overload khatarnak.

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Motor vs Generator + kaunsa Fleming rule

Q: Electric motor aur generator me kya antar? Dono me kaunsa Fleming rule lagta aur energy conversion kya?

Solution:

- MOTOR: electrical $\rightarrow$ mechanical energy. Field me current wali coil par force lagta $\rightarrow$ coil ghoomti. Fleming LEFT hand rule. Split ring/commutator use.
 - GENERATOR: mechanical $\rightarrow$ electrical energy. Coil ghumane par EMI se current. Fleming RIGHT hand rule. AC me slip ring, DC me split ring.
- Yaad: motor=LEFT, generator=RIGHT.

Example 2 - Fleming left hand apply karna

Q: Ek taar me current paper ke andar (x) ja rahi, magnetic field neeche se upar. Force ki direction?

Solution:

- Fleming left hand: first finger (field) upar, centre finger (current) andar, thumb (force) jis taraf aaye = force direction. Yahan thumb ek side (left/right) point karega - i.e. force taar ke perpendicular (horizontal).
- (Concept: F field aur current dono ke perpendicular hota.)

Example 3 - Electromagnetic induction

Q: Bina battery ke coil me current kaise paida kar sakte? Kis cheez par depend?

Solution:

- Magnet ko coil ke paas/door hilao (ya coil ko hilao) $\rightarrow$ magnetic field change $\rightarrow$ EMI se induced current. Galvanometer ye dikha deta.
- Depend: movement ki speed (jitni tez, utni jyada current) + coil ke turns.

Example 4 - Field lines property

Q: Do magnetic field lines kabhi cross kyun nahi karti?

Solution:

- Agar cross karein to us point par field ki 2 alag directions hongy, jo impossible hai (ek point par sirf ek direction). Isliye cross nahi karti.

Example 5 (EASIEST) - Earth wire ka kaam

Q: Ghar me earth (green) wire ka kya kaam hai?

Solution:

- Safety. Leakage/extra current ko zameen me bhej deta -> shock se bachata.

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Magnetic field lines ki 2 properties.
- Q2. Right hand thumb rule kya batata hai?
- Q3. Electromagnet me soft iron core kyun?
- Q4. AC aur DC me 1 antar.

MEDIUM (2-3 marks)

- Q5. Fleming left hand rule + kis device me use.
- Q6. Solenoid ka field bar magnet jaisa - samjhao.
- Q7. Domestic circuit ke 3 taar + kaam.
- Q8. Short circuit aur overloading me antar.

HARD (3-5 marks, board favourite)

- Q9. Electric motor ki working + split ring ka kaam.
- Q10. Electromagnetic induction + Faraday experiment + Fleming right hand.
- Q11. Generator working (AC vs DC, slip ring vs split ring).
- Q12. Earthing kya hai aur insaan ko shock se kaise bachata + fuse ka role.

ANSWER HINTS (PART D)

- A1. N se S bahar (closed loops); cross nahi karti; paas=strong field.
- A2. Current wale seedhe taar ke around field ki direction.
- A3. Jaldi magnetise + demagnetise hota (temporary strong magnet).
- A4. DC ek disha; AC disha badalti (ghar 50 Hz).
- A5. First=field, centre=current, thumb=force; MOTOR me use.
- A6. Ek sira N, doosra S, andar uniform field - bar magnet jaisa.
- A7. Live (current laata), neutral (wapas), earth (safety).
- A8. Short: live-neutral touch, achanak current; overload: bahut appliance.
- A9. Field me current coil par force (Fleming left) -> ghoomti; split ring current direction palat kar ek disha me ghumata.
- A10. Changing field se induced current; magnet-coil hilao; Fleming right hand.
- A11. Coil ghumao -> EMI; AC slip ring, DC split ring.
- A12. Metal body earth se jodna; leakage current zameen me; fuse zyada current par pighal kar bachata.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

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